WORKED SOLUTION

STEP I 2001, Question 3

Pure • Cubics and Discriminants

STEP I 2001, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q3

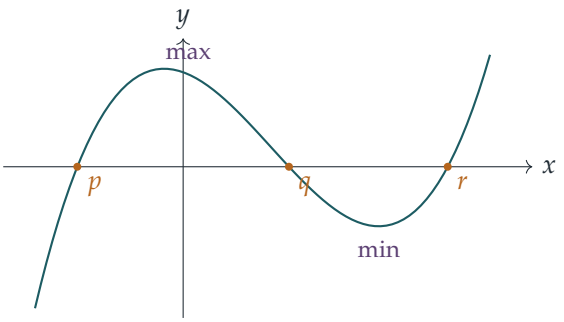
The key idea

A cubic with three distinct real roots must turn round between each pair of neighbouring roots, so its derivative, a quadratic, has two distinct real roots. Writing "two distinct real roots" as "discriminant positive" gives the first inequality. For the second part, the given product is a cubic whose coefficients are exactly the quantities in the target inequality, so the first part (or a completed square) does the work.

Solution

The sketch

Expanding, $f(x) = x^3 - (p+q+r)x^2 + \ldots$, so $f(x) \to +\infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$. The graph crosses the x-axis at p, q and r only, and since each root is simple, f changes sign at each: $f < 0$ for $x < p$, $f > 0$ on (p,q), $f < 0$ on (q,r) and $f > 0$ for $x > r$. So there is a local maximum somewhere between p and q and a local minimum somewhere between q and r. The y-intercept is $f(0) = -pqr$.



(The picture is drawn with $p < 0 < q < r$; the shape is the same wherever the roots lie, only the position of the y-axis changes.)

The first inequality

Expanding fully,

$$f(x) = x^3 - (p+q+r)x^2 + (qr+rp+pq)x - pqr,$$

so

$$f'(x) = 3x^2 - 2(p+q+r)x + (qr+rp+pq).$$

Since $f(p) = f(q) = 0$, Rolle's theorem gives a root of f' strictly between p and q; likewise there is a root strictly between q and r. These intervals don't overlap, so the quadratic $f'(x) = 0$ has two *distinct* real roots. (Without Rolle: $f'(p) = (p-q)(p-r) > 0$, $f'(q) = (q-p)(q-r) < 0$ and $f'(r) = (r-p)(r-q) > 0$, so f' changes sign on each interval.)

A quadratic has two distinct real roots exactly when its discriminant is positive, so

$$4(p+q+r)^2 - 12(qr+rp+pq) > 0, \qquad \text{that is,} \qquad (p+q+r)^2 > 3(qr+rp+pq),$$

as required.

The second inequality

Expanding,

$$(x^2+gx+h)(x-k) = x^3 + (g-k)x^2 + (h-gk)x - hk.$$

Compare this with the expansion of f: if the cubic has roots p, q, r, then

$$p+q+r = -(g-k), \qquad qr+rp+pq = h-gk,$$

so the target inequality $(g-k)^2 > 3(h-gk)$ is exactly the first inequality for the roots of this cubic.

Sufficient. Suppose $g^2 > 4h$. Then x^2+gx+h has two distinct real roots $\alpha < \beta$, and the cubic has real roots α, β, k. If k is different from both, the three roots are distinct and the first part gives $(g-k)^2 > 3(h-gk)$ at once.

Ask yourself What if k happens to equal α or β, so the roots are not distinct?

The first part assumed $p < q < r$, so we need a little more. The identity

$$(p+q+r)^2 - 3(qr+rp+pq) = \tfrac{1}{2}[(p-q)^2 + (q-r)^2 + (r-p)^2]$$

shows the left-hand side is positive unless $p = q = r$. Since $\alpha \neq \beta$, the three roots are never all equal, so the inequality holds in this case too.

A direct route avoids roots altogether. Multiplying out,

$$(g-k)^2 - 3(h-gk) = g^2 + gk + k^2 - 3h = (k + \tfrac{1}{2}g)^2 + \tfrac{3}{4}(g^2 - 4h),$$

which is a square plus a positive quantity when $g^2 > 4h$, and so positive. Either way, $g^2 > 4h$ is sufficient.

Not necessary. We need values with $g^2 \leq 4h$ for which the inequality still holds. Take $g = 0$, $h = 1$, $k = 2$. Then $g^2 = 0 \leq 4 = 4h$, but

$$(g-k)^2 = 4 > 3 = 3(h-gk).$$

So the inequality can hold without $g^2 > 4h$, and the condition is not necessary.

Ask yourself How can that be, when the first part needed three real roots?

The first part needed only that $f'(x) = 0$ has two distinct real roots, which is weaker. Here the cubic $(x^2+1)(x-2) = x^3 - 2x^2 + x - 2$ has only one real root, but $f'(x) = 3x^2 - 4x + 1 = (3x-1)(x-1)$, so it still has two stationary points. The inequality $(g-k)^2 > 3(h-gk)$ says precisely that the cubic has a distinct local maximum and minimum, whether or not it crosses the axis three times.

Checks

- With $p, q, r = 0, 1, 2$: $(p+q+r)^2 = 9 > 6 = 3(qr+rp+pq)$, and $f'(x) = 3x^2 - 6x + 2$ has roots $1 \pm 1/\sqrt{3}$, one in $(0,1)$ and one in $(1,2)$.
- The completed-square formula shows the boundary of the inequality is sharp: with $g = 2$, $h = 1$ (so $g^2 = 4h$) and $k = -1$ we get $(g-k)^2 = 9 = 3(h-gk)$, equality and not strict inequality. This is the cubic $(x+1)^3$, with all three roots equal.
- A computer checked both identities and both inequalities exactly on 20 000 random rational cases each, plus the example $g = 0$, $h = 1$, $k = 2$ (giving $4 > 3$).

Common slips

- Sketching the cubic the wrong way up: the coefficient of x^3 is $+1$, so the graph rises to the right.
- Forgetting the case where k is a root of $x^2 + gx + h$, where the first part does not apply as stated.
- Showing only that the inequality holds for one example with $g^2 \leq 4h$ and calling that "sufficient". One example proves *not necessary*; sufficiency needs a proof for all g, h, k.

Beyond STEP

The quantity $(p+q+r)^2 - 3(qr+rp+pq)$ is, up to a factor, the discriminant of the derivative, and it is a *symmetric function* of the roots: any such function can be written in terms of the coefficients, which is why the inequality could be translated from roots to g, h, k. The full discriminant of a cubic, $\prod(p-q)^2$, decides whether all three roots are real; this question uses its weaker cousin. Symmetric functions and discriminants lead on to Galois theory; Allenby's *Rings, Fields and Groups* is a good place to start.

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